



AI Driven SQA: Manual & Automation Testing

Assignment 03

Module: 03

Topic: Software Testing Life Cycle (STLC) with AI

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Q1. Create a Test Plan for an Online Shopping Website?

Answer:

Software Test Plan Document.

Project Name: Online Shopping Website.

Document Title: Test Plan for Online Shopping Website.

Prepared By: Jhion Adbar Opi.

The purpose of this Test Plan is to define the testing approach, scope, objectives, resources, schedule and activities required to validate the functionality and quality of the Online Shopping Website.

The testing process will focus on verifying the major modules of the system, including:

01.Login Checklist

02.Registration Checklist

03.Checkout Checklist

Test Objectives

The main objectives of testing are:

To verify that all functional requirements of the website are implemented correctly.

- To ensure users can successfully register and authenticate into the system.
- To validate product selection, cart management and checkout workflow.
- To verify that payment and order confirmation processes work correctly.
- To ensure the application provides a secure and user-friendly experience.
- To verify system compatibility across different browsers and devices.

Test Scope

The following features will be tested:

Module	Testing Activities
User Registration	Validate account creation, input fields, password rules, and duplicate account handling
User Login	Verify authentication, invalid credentials handling, session management
Product Search	Validate searching, filtering, and product availability
Product Details	Verify product information, price, images, and description
Shopping Cart	Test adding, updating, and removing products
Checkout	Validate customer information, shipping address, order summary
Payment	Verify payment processing and transaction validation
Order Management	Verify order confirmation and order tracking

The following areas are excluded from this testing process:

Feature	Reason
Third-party payment provider internal system	Managed by external payment service provider
Server hardware testing	Managed by infrastructure team
Database performance tuning	Outside application testing scope
Production monitoring	Performed after deployment

Test Strategy

The following testing methods will be performed:

Testing Type	Purpose
Functional Testing	Verify application functions according to requirements
UI Testing	Validate user interface elements and usability
Integration Testing	Verify communication between different modules
Regression Testing	Ensure new changes do not break existing features
Security Testing	Verify authentication and data protection
Performance Testing	Validate system response time and stability
Compatibility Testing	Verify functionality across browsers and devices

Test Approach

The testing process will include the following activities:

1. Requirement Analysis; QA team will analyze business requirements and identify test conditions.
2. Test Planning: Testing scope, resources, tools, and schedule will be defined.
3. Test Case Design: Detailed test cases and test data will be prepared.
4. Test Execution: Test cases will be executed and actual results will be compared with expected results.
5. Defect Reporting: Detected issues will be recorded, prioritized, and communicated to the development team.
6. Test Closure: Final test reports and quality assessment will be prepared.

Defect Management Process

Defect information will include:

Field	Description
Defect ID	Unique defect number
Summary	Short description of issue
Severity	Impact level of defect
Priority	Fix urgency
Steps to Reproduce	Steps for recreating issue
Status	Current defect state
Assigned To	Responsible developer

Risk Management

Risk	Impact	Mitigation Plan
Requirement changes	Test cases may become outdated	Update requirements and test cases regularly
Environment failure	Testing delay	Maintain backup environment
Payment integration issues	Checkout testing blocked	Use sandbox payment system
Large number of defects	Release delay	Prioritize critical defects first

Test Metrics

The following metrics will be measured:

Metric	Purpose
Test Case Execution Rate	Measure testing progress
Test Pass Percentage	Measure successful tests
Defect Density	Measure software quality
Defect Resolution Rate	Measure fixing efficiency
Test Coverage	Measure requirement validation

Q2. Prepare a Test Checklist.

Answer:

01. Login Module Checklist

To validate user authentication functionality and ensure registered users can securely access their accounts.

Test Area	Validation Point	Expected Result
Login Page	Verify login page accessibility	Login page should load successfully
User Interface	Verify username/email field visibility	Username/email field should be displayed properly
User Interface	Verify password field visibility	Password field should be displayed properly
Input Validation	Verify user can enter valid email address	Valid email should be accepted
Input Validation	Verify invalid email format validation	System should display proper validation message
Password Validation	Verify password field hides entered characters	Password should be masked for security
Mandatory Field Validation	Verify login with empty username	System should display username required message
Mandatory Field Validation	Verify login with empty password	System should display password required message
Authentication	Verify login using valid credentials	User should successfully login and access account dashboard
Authentication	Verify login using incorrect password	System should display invalid credential message
Authentication	Verify login using unregistered email	System should prevent login and show error message
Security	Verify multiple failed login attempts	System should apply security control such as account lock or warning
Session Management	Verify user logout functionality	User should be logged out successfully
Password Recovery	Verify forgot password functionality	User should navigate to password reset page

Fig 2.1: Login Module Checklist.

Module 1: Login Test Cases

Requirement ID	Requirement	Module	Test Case ID	Test Case	Test Data	Expected Result	Actual Result	Status	Priority	Severity	Note
REQ_LOGIN_001	System should provide user login functionality	Login	TC_LOGIN_001	Verify login page accessibility	Username: standard_user / Password: secret_sauce	Login page should load successfully	Work as expected	Pass	High	Medium	Basic availability test show valid.
REQ_LOGIN_002	System should provide username/email input field	Login	TC_LOGIN_002	Verify username field visibility	Username: problem_user / Password: secret_sauce	Username/email field should be displayed	Work as expected	Pass	High	Medium	UI validation valid.
REQ_LOGIN_003	System should provide password input field	Login	TC_LOGIN_003	Verify password field visibility	Username: problem_user / Password: secret_sauce	Password field should be displayed	Work as expected	Pass	High	Medium	UI validation valid
REQ_LOGIN_004	System should accept valid email format	Login	TC_LOGIN_004	Verify valid email input	standard_user / problem_user / performance_glitch_user / error_user / visual_user, each with Password: secret_sauce	Email should be accepted	It show moderate usability.	Not Executed	High	Medium	Input validation need to be perfect.
REQ_LOGIN_005	System should validate email format	Login	TC_LOGIN_005	Verify invalid email validation	Password: secret_sauce Username: ' standard_user ' / Password: secret_sauce	Error message should display	Not Executed	Fail	Medium	Medium	Negative testing not valid.
REQ_LOGIN_006	System should protect user password	Login	TC_LOGIN_006	Verify password masking	Password: secret_sauce	Password characters should be hidden	Not Executed	Not Executed	Medium	Low	Security validation not valid.
REQ_LOGIN_007	System should validate mandatory login fields	Login	TC_LOGIN_007	Login without username	Password: Test@123	Username required message should display	Not Executed	Not Executed	High	Medium	Required field validation not valid.
REQ_LOGIN_008	System should validate mandatory login fields	Login	TC_LOGIN_008	Login without password	Email:user@test.com Username: invalid_user / Password: secret_sauce	Password required message should display	It do show as expected	Pass	High	Medium	Required field validation valid
REQ_LOGIN_009	System should authenticate registered users	Login	TC_LOGIN_009	Login with valid credentials	user@test.com / Test@123 standard_user / problem_user / performance_glitch_user / error_user / visual_user, each with Password: secret_sauce	User should login successfully	Not Executed	Not Executed	Critical	Critical	Positive scenario not valid
REQ_LOGIN_010	System should prevent unauthorized login	Login	TC_LOGIN_010	Login with wrong password	Username: invalid_user / Password: secret_sauce	Invalid credential message should appear	Not Executed	Not Executed	High	High	Security validation not valid
REQ_LOGIN_011	System should prevent login from unknown account	Login	TC_LOGIN_011	Login using unregistered email	Username: invalid_user / Password: secret_sauce	User should not be authenticated	Not Executed	Fail	High	High	Authentication test fail
REQ_LOGIN_012	System should protect against repeated failed attempts	Login	TC_LOGIN_012	Verify multiple failed login attempts	Wrong password ×5 Username: ' standard_user ' / Password: secret_sauce	Account lock/warning should appear	Fail	Fail	High	Critical	Security feature totally fail
REQ_LOGIN_013	System should allow users to logout	Login	TC_LOGIN_013	Verify logout functionality	Username: standard_user / Password: Sauce@123	User should logout successfully	Glitch	Fail	Medium	High	Session validation fail
REQ_LOGIN_014	System should provide password recovery	Login	TC_LOGIN_014	Verify forgot password option	Registered email Username: standard_user / Password: 'secret sauce'	Reset password page should open	Not Executed	Fail	Medium	Medium	Recovery workflow fail

Fig 2.2: Login Module Test Case.

02. Registration Module Checklist

To verify that new customers can create accounts successfully with valid information.

Test Area	Validation Point	Expected Result
Registration Page	Verify registration page accessibility	Registration page should open successfully
User Information	Verify name field availability	Name field should be displayed
User Information	Verify user can enter valid name	Name should be accepted
Email Validation	Verify registration using valid email	Account creation should proceed
Email Validation	Verify duplicate email registration	System should reject existing email
Password Validation	Verify password field functionality	Password should accept valid input
Password Validation	Verify weak password validation	System should display password strength warning
Password Validation	Verify confirm password matching	Matching password should be accepted
Password Validation	Verify password mismatch validation	Error message should appear
Phone Validation	Verify valid phone number input	Phone number should be accepted
Phone Validation	Verify invalid phone number format	Validation message should appear
Mandatory Fields	Verify registration with empty fields	Required field messages should display
Terms & Conditions	Verify agreement checkbox validation	User must accept terms before registration
Security Validation	Verify CAPTCHA validation	Incorrect CAPTCHA should prevent registration
Account Creation	Verify successful account registration	New account should be created successfully

Fig 2.3: Registration Module Checklist.

Module 2: Registration Test Cases

Requirement ID	Requirement	Module	Test Case ID	Test Case	Test Data	Expected Result	Actual Result	Status	Priority	Severity	Note
REQ_REG_001	User should create a new account	Registration	TC_REG_001	Open registration page	Registration URL	Registration page should load	Work as expected	Pass	High	Medium	Page validation valid
REQ_REG_002	System should accept user name	Registration	TC_REG_002	Enter valid name	Valid integer	Name should be accepted	Work as expected	Pass	Medium	Low	Input validation valid
REQ_REG_003	System should accept valid email	Registration	TC_REG_003	Register using valid email	jhionadbar@test.com	Email should be accepted	Work as expected	Pass	High	Medium	Email validation valid
REQ_REG_004	System should prevent duplicate accounts	Registration	TC_REG_004	Register with existing email	opi@test.com	Duplicate email error should appear	Not Executed	Fail	High	High	Database validation fail
REQ_REG_005	System should allow password creation	Registration	TC_REG_005	Enter valid password	standard_user / problem_user / performance_glitch_user / error_user / visual_user, each with Password: secret_sauce	Password should be accepted	Password do accepted	Pass	High	Medium	Password validation valid
REQ_REG_006	System should enforce password strength	Registration	TC_REG_006	Enter weak password	123456789	Password strength warning should appear	Not Executed	Fail	Medium	Medium	Security validation fail
REQ_REG_007	System should verify confirm password	Registration	TC_REG_007	Enter matching passwords	Test@123/Test@123	Registration should continue	Executed	Not Executed	High	Medium	Validation fail
REQ_REG_008	System should reject mismatched passwords	Registration	TC_REG_008	Enter different passwords	Test123/Test456	Error message should display	Not Executed	Fail	High	Medium	Negative scenario fail
REQ_REG_009	System should validate phone number	Registration	TC_REG_009	Enter valid phone number	01*****	Phone number accepted	Not Executed	Not Executed	Medium	Low	Input validation
REQ_REG_010	System should reject invalid phone format	Registration	TC_REG_010	Enter invalid phone	01*****	Validation message displayed	Not Executed	Not Executed	Medium	Medium	Data validation
REQ_REG_011	System should validate required fields	Registration	TC_REG_011	Submit empty form	Blank fields	Required messages displayed	Not Executed	Not Executed	High	Medium	Mandatory validation
REQ_REG_012	User must accept terms and conditions	Registration	TC_REG_012	Register without accepting terms	Checkbox unchecked. Username: ' OR '1'='1 / Password: ' OR '1'='1	Registration should not proceed	Not Executed	Not Executed	Medium	Medium	Business rule
REQ_REG_013	System should validate CAPTCHA	Registration	TC_REG_013	Enter incorrect CAPTCHA	Wrong CAPTCHA	Registration should fail	Not Executed	Fail	Medium	Medium	Security testing fail
REQ_REG_014	System should create account successfully	Registration	TC_REG_014	Complete valid registration	Valid user integer data.	Account should be created	Not Executed	Fail	Critical	Critical	Main workflow fails
REQ_REG_015	System should verify registered email	Registration	TC_REG_015	Check verification email	Registered email	Verification email should arrive	Not Executed	Not Executed	Medium	High	Email workflow

Fig 2.4: Registration Test Cases

03. Checkout Module Checklist

To verify that customers can complete the purchase process successfully.

Test Area	Validation Point	Expected Result
Shopping Cart	Verify selected product appears in cart	Product should display correctly
Shopping Cart	Verify product quantity update	Quantity should update successfully
Shopping Cart	Verify product removal from cart	Product should be removed successfully
Price Calculation	Verify product price display	Correct product price should appear
Price Calculation	Verify total amount calculation	Total price should be calculated correctly
Discount	Verify valid coupon application	Discount should apply successfully
Discount	Verify invalid coupon handling	Proper error message should display
Customer Information	Verify shipping address entry	Address should be accepted
Customer Information	Verify mandatory address fields	Required validation should appear
Shipping	Verify available shipping methods	Shipping options should display
Shipping	Verify shipping method selection	Selected shipping method should save
Payment	Verify payment page loading	Payment page should open successfully
Payment	Verify valid payment information	Payment should complete successfully
Payment	Verify invalid payment information	Error message should appear
Order Confirmation	Verify order placement	Order should be created successfully
Order Confirmation	Verify order ID generation	Unique order ID should be generated
Order Notification	Verify confirmation email/message	Customer should receive order confirmation

Fig 2.5: Checkout Module Checklist

Module 3: Checkout Test Cases

Requirement ID	Requirement	Module	Test Case ID	Test Case	Test Data	Expected Result	Actual Result	Status	Priority	Severity	Note
REQ_CART_001	User should manage shopping cart	Checkout	TC_CART_001	Verify product appears in cart	Products	Product should display correctly	Display correctly	Pass	High	High	Cart validation valid and pass
REQ_CART_002	User should update product quantity	Checkout	TC_CART_002	Change product quantity	Quantity: 1++	Quantity should update	Not Executed	Fail	Medium	Medium	Cart operation fail
REQ_CART_003	User should remove product	Checkout	TC_CART_003	Remove item from cart	Products	Product should be removed	Not Executed	Fail	Medium	Medium	Cart operation fail
REQ_CHECK_001	System should calculate product price	Checkout	TC_CHECK_001	Verify product price	Product: TK 1++	Correct price displayed	Not Executed	Fail	High	High	Calculation fail
REQ_CHECK_002	System should calculate total amount	Checkout	TC_CHECK_002	Verify total calculation	Product: TK 1 ++ Shipping: TK 1++	Total should show TK (1++)	Not Executed	Not Executed	High	High	Calculation no data
REQ_CHECK_003	System should apply discount coupon	Checkout	TC_CHECK_003	Apply valid coupon	CODE	Discount should apply	Not Executed	Not Executed	Medium	Medium	Promotion testing no data
REQ_CHECK_004	System should reject invalid coupon	Checkout	TC_CHECK_004	Apply invalid coupon	INVALID CODE	Error message should display	Wrong Massage	Fail	Medium	Medium	Negative testing fail
REQ_CHECK_005	User should provide shipping address	Checkout	TC_CHECK_005	Enter delivery address	Place, Bangladesh	Address should save	Wrong Address	Fail	High	High	Shipping validation fail
REQ_CHECK_006	System should validate mandatory address fields	Checkout	TC_CHECK_006	Submit incomplete address	Missing postcode	Required message displayed	Not Executed	Not Executed	Medium	Medium	Validation no data
REQ_CHECK_007	System should provide shipping options	Checkout	TC_CHECK_007	View shipping methods	Methods should be shown .	Available options displayed	Not Executed	Fail	Medium	Low	UI validation fail
REQ_CHECK_008	User should select shipping method	Checkout	TC_CHECK_008	Select delivery option	Express Delivery	Selected option saved	Glitch	Fail	Medium	Medium	Workflow validation sometimes fail
REQ_PAY_001	System should process payment	Payment	TC_PAY_001	Make successful payment	Valid test card	Payment completed successfully	Payment go through	Pass	Critical	Critical	Payment validation pass
REQ_PAY_002	System should handle payment failure	Payment	TC_PAY_002	Use invalid payment details	Invalid card	Error message displayed	Not Executed	Not Executed	Critical	Critical	Negative payment test data corruption
REQ_ORDER_001	System should create order	Checkout	TC_ORDER_001	Place order	Valid checkout data	Order should be created	Not Executed	Not Executed	Critical	Critical	Main business flow unknown
REQ_ORDER_002	System should generate order ID	Checkout	TC_ORDER_002	Verify order confirmation	Completed order	Unique order ID generated	List is shown	Pass	High	High	Order tracking valid and pass
REQ_ORDER_003	System should notify customer	Checkout	TC_ORDER_003	Verify order confirmation email	Customer email	Confirmation message/email received	Not Executed	Not Executed	Medium	Medium	Notification testing system fail.

Fig 2.6: Module 3: Checkout Test Case

